VEHICLE TRACKING + OCCLUSIONS

Phase 1

Overall Summary and Continuity Guide

Master handoff for Phase 1A through Phase 1I and their detailed documentation

Status	COMPLETE / FROZEN
Pilot basis	12 frozen UA-DETRAC events E001-E012; SAM2.1 Tiny and DAM4SAM Tiny
Purpose	Restore the full Phase 1 logic quickly, then point to the detailed phase PDF when exact evidence is needed.
Central result	The target was usually available as a candidate after verified return; the main recoverable bottleneck became identity selection and genuine ambiguity.
Boundary	Final sequential acquisition of plate/color/make-model/local cues is not part of Phase 1.

Use this as the first context document

The detailed Phase 1 PDFs remain authoritative. This summary tells another chat what each phase asked, what it built, what was frozen, the safest numbers to carry forward, and exactly which detailed PDF to open for deeper evidence.

Consolidated state: 06 October 2026

1. How to Use This Summary

Read this document for continuity and system-level reasoning. Open a detailed phase PDF for implementation details, event-level evidence, intermediate corrections, figures, exact commands, thresholds, or limitations. If a compressed statement here seems broader than a detailed phase record, the detailed record controls.

Evidence rule

Measured = evaluator/CSV/terminal output. Visually reviewed = original-frame or overlay inspection. Frozen = accepted Phase 1 baseline decision. Pilot/exploratory = useful design evidence, not a validated generalization claim.

1.1 Detailed documentation map

Phase	Detailed record scope	Filename	pp.
1A-1C	Literature-guided tracker planning; framework; SAM2.1/DAM4SAM baseline runs	Vehicle_Tracking_Phase_1A_1B_1C_Research_Documentation.pdf	15
1D-1E	Representative validation status; offline event-aware loss/recovery reference	Vehicle_Tracking_Phase_1D_1E_Research_Documentation.pdf	15
1F	L1 simple loss detector and L2 Sentinel-inspired reliability states	Vehicle_Tracking_Phase_1F_Research_Documentation.pdf	14
1G	Conditional YOLO road-vehicle candidate generation after risk/loss	Vehicle_Tracking_Phase_1G_Research_Documentation.pdf	12
1H	Appearance identity, M1/M2 memory, ambiguity, observation quality, selective routing	Vehicle_Tracking_Phase_1H_Research_Documentation.pdf	16
1I	F1-F8 failure taxonomy, manual reviews, final primary/secondary assignments	Vehicle_Tracking_Phase_1I_Research_Documentation.pdf	14

1.2 Reading shortcut

- Need baseline tracker setup or literature rationale -> open 1A-1C.
- Need the offline reference definition of loss/recovery -> open 1D-1E.
- Need persistent CRITICAL/LOST logic -> open 1F.
- Need detector sweep, filter, candidate counts, runtime -> open 1G.
- Need identity scores, memory experiments, R1/R2 -> open 1H.
- Need final F1-F8 assignments -> open 1I.

2. Phase 1 Role in the Research Project

Working title: Reliability-Aware Sequential Identity Evidence Acquisition for Post-Occlusion Recovery of User-Selected Road Vehicles.

Primary research question

Can reliability-aware sequential acquisition of vehicle identity cues improve post-occlusion recovery of a user-selected road motor vehicle among visually similar distractors, while reducing false identity decisions and average inference cost compared with always-on multimodal Re-ID and fixed coarse-to-fine recovery?

Phase 1 did not implement the final sequential cue policy. It built and diagnosed the prerequisite pipeline so later methodology is driven by observed failures rather than assumptions.

2.1 End-to-end Phase 1 chain

1A-C	1D-E	1F	1G	1H	1I
Trackers	Offline reference	Online risk	Candidates	Identity / defer	Taxonomy

2.2 Inherited Phase 0 basis

All Phase 1 experiments use the frozen, manually reviewed Phase 0 UA-DETRAC pilot inventory E001-E012. Phase 1 does not re-mine events. Phase 0 remains the authority for event construction, selection frames, occlusion/reappearance anchors, and scenario intent.

Pilot limitation

E005-E007 reuse overlapping footage of the same stationary SUV and are not independent hard moving-target examples. Phase 1 is baseline/design evidence, not a final benchmark.

Phase 1A-1C	Literature-guided design, common framework, and tracker baselines
FROZEN	**Question:** Can the project translate the literature into a controlled baseline pipeline and run standard versus distractor-aware trackers on the frozen events? **Detailed PDF:** Vehicle_Tracking_Phase_1A_1B_1C_Research_Documentation.pdf

Literature-guided design, common framework, and tracker baselines

3.1 Phase 1A - literature to baseline design

VeTrack, Sentinel, DAM4SAM, and DeconfuseTrack were read for implementation lessons. The phase used literature to select strong baselines and define what should be measured; it did not freeze the final novelty or cue-acquisition equations.

- VeTrack -> quality/confidence-aware association ideas.
- Sentinel -> explicit reliability/loss states.
- DAM4SAM -> distractor-aware SAM2.1 baseline and cautious memory updates.
- DeconfuseTrack -> overlapping candidates should not be suppressed without evidence.

3.2 Phase 1B - common infrastructure

Implemented the frozen event loader, tracker interface, runner, and evaluator. A static debug tracker first verified the plumbing. Key conceptual correction: XML annotation presence or continuous tracker output is not, by itself, proof of visible post-occlusion identity recovery.

3.3 Phase 1C - baseline execution

Trackers	SAM2.1 Tiny and DAM4SAM Tiny
Scope	All 12 frozen events
Runtime	Both about 1.1 s/frame locally; SAM2.1 faster in 11/12 event-local averages
Hard-case observation	DAM4SAM was more cautious in E006/E012 where SAM2 showed false continuation/drift-like behavior
Need exposed	A recovery-aware evaluator was required because naive continuity metrics mixed easy tracking, occlusion, loss, and identity errors.

Carry-forward

Phase 1C established real baseline trajectories but deliberately did not call all continued boxes “recovery.” That motivated the event-aware offline reference in 1D-1E.

Phase 1D-1E	Representative validation and recovery-aware offline evaluation
FROZEN	**Question:** Can the tracker outputs be interpreted with correct event-aware loss/recovery semantics? **Detailed PDF:** Vehicle_Tracking_Phase_1D_1E_Research_Documentation.pdf

Representative validation and recovery-aware offline evaluation

4.1 Phase 1D status

The planned representative-event validation had already been satisfied by the representative-first execution and manual review in Phase 1C, so no duplicate rerun was required.

4.2 Phase 1E - offline reference layer

Phase 1E separated continuous tracking, genuine loss, verified reappearance, automatic recovery, possible switch warnings, and candidate retention. It is a GT-assisted evaluation layer, not a deployable detector.

Evaluation unit	12 events x 2 trackers = 24 runs
Reference loss	8/24 runs; 16/24 without true loss
Event pattern	E005-E007: genuine loss/resumption; E012: loss with no verified return
Automatic recovery	Three loss events had verified reappearance; E012 did not
Reference candidate retention	Annotation-backed debug candidates retained target in 10/10 eligible recorded-reappearance events

Frozen role

Phase 1E became the offline reference against which online loss detection and later recovery components were judged.

Phase 1F	Online loss detection baselines
FROZEN	**Question:** Can the running system decide that tracking is unreliable or lost without using target ground truth? **Detailed PDF:** Vehicle_Tracking_Phase_1F_Research_Documentation.pdf

Online loss detection baselines

5.1 L1 - simple conservative rule

L1 used runtime presence/geometry only. Missing/invalid output or a combination of very low temporal IoU and a large normalized center jump was unreliable; three consecutive unreliable frames produced sticky LOST.

L1 result	TP 7, TN 16, FP 0, FN 1
Mean timing offset	about +6.71 frames
Main miss	E012 SAM2 smooth drift/false continuation

5.2 L2 - Sentinel-inspired reliability states

L2 used CONFIDENT, UNCERTAIN, CRITICAL, and LOST. Because saved tracker confidence fields were empty, this is Sentinel-inspired rather than an exact reproduction.

Risk-trigger result	TP 8, TN 14, FP 2, FN 0
Mean timing offset	about +2.62 frames
False triggers	E004 for both trackers
Frozen activation	persistent CRITICAL OR LOST; transient Critical alone does not trigger

Frozen decision

Carry persistent CRITICAL OR LOST into Phase 1G. This accepts two false activations to catch the smooth E012 failure that L1 missed.

Phase 1G	Conditional real vehicle candidate generation
FROZEN	**Question:** After a reliability trigger, can a lightweight real detector propose plausible vehicles without dropping the true target? **Detailed PDF:** Vehicle_Tracking_Phase_1G_Research_Documentation.pdf

Conditional real vehicle candidate generation

6.1 Frozen candidate generator

Detector	YOLO11n; car/motorcycle/bus/truck
Operating point	imgsz 640; confidence 0.20
Broad filter	area-ratio [0.35,15], aspect-ratio ratio [0.5,2] relative to original selection box
Identity decision	None - Phase 1G only proposes candidates

6.2 Final pilot results

Target retention	10/10 eligible recorded-reappearance events before and after filtering
Candidate pool	179 raw -> 146 filtered; 18.4% reduction
Mean filtered set	14.6 candidates/event
Triggered eligible runs	8/8 contained the target at first valid recovery-scoring frame
Genuine loss/reappearance runs	6/6 for E005-E007 across both trackers
Conditional cost	975 detector frames after triggers; 104 caused by E004 false triggers

6.3 Interpretation

A proposal delay of 0 means the target was available at the first frame where verified reappearance scoring became valid; it does not mean the detector independently found physical visibility onset.

Frozen conclusion

Candidate availability was not the dominant pilot bottleneck. Phase 1G handed a contained target plus distractors to Phase 1H for identity reasoning.

Phase 1H	Identity matching, memory, ambiguity, and selective routing
FROZEN	**Question:** If the target is in the candidate set, can appearance identify it safely - or recognize when more evidence is required? **Detailed PDF:** Vehicle_Tracking_Phase_1H_Research_Documentation.pdf

Identity matching, memory, ambiguity, and selective routing

7.1 Appearance baseline and positive memory

Identity evaluation used 10 verified-reappearance events, Phase 1G candidates, and an ImageNet-pretrained ResNet50 2048-D L2-normalized embedding. GT was used only after ranking.

H-0 single selection	Rank-1 5/10; Rank-3 6/10; mean rank 4.70; MRR 0.585
M1 recent positive memory	Rank-1 5/10; Rank-3 5/10; mean rank 4.40; MRR 0.574
M1 quality audit	25/26 strong recent entries, 1 weak, 0 contaminated

7.2 M2 distractor memory lesson

A runtime hard-distractor memory was built, but direct scoring as positive similarity minus distractor similarity was structurally harmful: the true target also resembles deliberately selected hard distractors.

Naive M2	Rank-1 4/10; Rank-3 6/10; mean rank 4.30; MRR 0.551
Failure diagnostic	In all 5 active M2 events, the true target had stronger positive evidence but also a larger negative-memory penalty

7.3 Selective routing

Top1-Top2 candidate margin caught most wrong identity choices; E004 was confidently wrong by appearance but its winning detection had very low within-frame detector-confidence percentile.

R1 always accept	10 accepted; 5 correct; 5 false matches
Exploratory R2	margin >= 0.03 AND confidence percentile >= 0.50
R2 pilot	5 accepted; 5/5 correct; 0 false accepted; 50% coverage; 5 NEED_MORE_EVIDENCE
Caution	R2 thresholds were derived on the same pilot and are not validated final thresholds

Frozen conclusion

Do not force identity. The preferred direction is MATCH when evidence is reliable, otherwise NEED MORE EVIDENCE. Phase 1H does not yet acquire the extra cue.

Phase 1I	Failure taxonomy and systematic error analysis
FROZEN	**Question:** What actually fails in the frozen pipeline, and which failure modes should drive later methodology? **Detailed PDF:** Vehicle_Tracking_Phase_1I_Research_Documentation.pdf

Failure taxonomy and systematic error analysis

8.1 Taxonomy contract

Phase 1I combined frozen tracker, offline reference, online trigger, candidate, identity, and manual-review evidence into primary plus optional secondary failure labels. F8 is an event condition; F3 is tracker-specific; F5/F7 are mainly identity-stage event findings.

8.2 Final unique-event findings

F1 - drift before occlusion	0 confirmed
F2 - target completely lost	4/12: E005,E006,E007,E012
F3 - jump to similar distractor	2/12: E006/SAM2,E012/SAM2
F4 - returned but not detected	0 among eligible verified returns
F5 - detected but not selected	5/12: E001,E003,E004,E008,E011
F6 - memory contamination	0 confirmed
F7 - genuine ambiguity	3/12: E001,E008,E011
F8 - never returned	1/12: E012
Unresolved	E010 deliberately unresolved

8.3 Research consequence

F5 is the main recoverable failure and F7 is a meaningful subset. Since F4 was not observed on the eligible return set, later work should focus on additional identity evidence rather than indiscriminately expanding detection.

Phase 1 terminal diagnosis

The pilot problem is now evidence-driven: reliable tracking/loss handling and candidate proposal are necessary, but appearance-only identity is insufficient when the view changes or similar vehicles remain plausible.

9. Consolidated Phase 1 Snapshot

Use this page as the fastest numeric restore point. All values are pilot results and must retain the limitations stated in the detailed phase records.

Phase	Layer	Unit	Frozen takeaway
1A-C	Baseline tracking	SAM2.1 + DAM4SAM on 12 events	Baseline trajectories established; naive box continuity was not sufficient evidence of recovery.
1D-E	Offline reference	24 tracker/event runs	8 reference-loss runs; E005-E007 genuine loss/resumption; E012 no verified return.
1F	Online trigger	L1 vs L2	L1 7/8 reference losses; L2 risk trigger 8/8 with 2 false E004 activations.
1G	Candidate proposal	10 eligible return events	Target retained 10/10; 179 -> 146 candidates; 14.6/event.
1H	Identity	10 identity events	H0 Rank-1 5/10; M1 5/10; naive M2 4/10; exploratory deferral removed false accepted matches on pilot.
1I	Diagnosis	12 events / 24 rows	F5 in 5 events; F7 in 3; F2 in 4; F3 in 2; F4/F6 unconfirmed; E010 unresolved.

9.1 End-of-Phase-1 evidence chain

1. **Tracking:** baseline can remain correct, lose the target, or jump.
2. **Reliability:** persistent Critical/Lost can trigger recovery without GT.
3. **Candidates:** verified returns were retained in the pilot candidate set.
4. **Identity:** appearance-only ranking fails in several similar/view-changed cases.
5. **Decision:** deferral is safer than forced identity when evidence is weak.
6. **Taxonomy:** F5/F7 empirically justify later selective acquisition of additional identity evidence.

10. What Phase 1 Established - and What It Did Not

10.1 Safe conclusions to carry forward

- Correct post-occlusion identity cannot be inferred from tracker continuity alone.
- Online reliability state can conditionally activate recovery processing.
- On the pilot, the true target was available in the candidate set whenever verified reappearance scoring was possible.
- The major recoverable bottleneck moved to identity selection among plausible candidates.
- Recent positive memory alone did not solve identity; naive hard-distractor subtraction was unsafe.
- The system should be allowed to output NEED MORE EVIDENCE instead of forcing a match.
- Observed F5/F7 failures provide the empirical bridge to later additional vehicle-identity cues.

10.2 Claims Phase 1 must not make

- The 12-event pilot is not a final benchmark or generalization result.
- 10/10 candidate retention is not 100% end-to-end recovery accuracy.
- XML annotation persistence is not proof of visual target visibility.
- L2 is Sentinel-inspired, not an exact Sentinel reproduction.
- Exploratory R2 thresholds are not validated because they were derived and evaluated on the same small pilot.
- E005-E007 are overlapping observations of the same stationary SUV, not independent hard cases.
- E010 must remain unresolved unless new evidence is produced.

Phase 1 freeze boundary

The final sequential acquisition of plate, color, make/model, local features, learned Re-ID, or other cues has not been implemented in Phase 1. Phase 1 ends after establishing when and why additional evidence is needed.

11. Frozen Handoff for a New Chat

A new chat should load these facts first and avoid reopening Phase 1 unless the user explicitly asks to revise it.

Research direction	Reliability-aware sequential identity evidence acquisition for post-occlusion recovery of a user-selected road vehicle among visually similar distractors.
Pilot	Phase 0 frozen UA-DETRAC E001-E012.
Trackers	SAM2.1 Tiny and DAM4SAM Tiny.
Online activation	Phase 1F L2: persistent CRITICAL OR LOST.
Candidate generator	Phase 1G YOLO11n, imgsz 640, conf 0.20, road-vehicle classes, broad geometry filter.
Identity baseline	Phase 1H ResNet50 H0/M1/M2 are frozen baseline evidence.
Decision direction	MATCH only when reliable; otherwise NEED MORE EVIDENCE. Thresholds are not yet validated.
Failure priority	F5 is the main recoverable failure; F7 is a meaningful subset; F4 candidate absence was not observed among eligible verified returns.
End state	Phase 1 COMPLETE / FROZEN.

11.1 Detailed-document routing

Baseline setup / literature / tracker runs: open Phase 1A-1C detailed PDF.

Offline loss/recovery reference: open Phase 1D-1E detailed PDF.

L1/L2 states and trigger: open Phase 1F detailed PDF.

Detector/filter/candidate metrics: open Phase 1G detailed PDF.

H0/M1/M2 and selective routing: open Phase 1H detailed PDF.

F1-F8 final taxonomy: open Phase 1I detailed PDF.

Terminal statement

Phase 1 converted the original idea into an evidence-driven systems diagnosis. The remaining research problem is reliable identity recovery when appearance is insufficient or multiple vehicles remain plausible. Later methodology should address that problem selectively rather than by blindly increasing tracking, detection, or multimodal compute.

Appendix A. Document and Artifact Index

A.1 Detailed Phase 1 PDFs

1A-1C: **Vehicle_Tracking_Phase_1A_1B_1C_Research_Documentation.pdf** (15 pages) - Literature-guided tracker planning; framework; SAM2.1/DAM4SAM baseline runs
1D-1E: **Vehicle_Tracking_Phase_1D_1E_Research_Documentation.pdf** (15 pages) - Representative validation status; offline event-aware loss/recovery reference
1F: **Vehicle_Tracking_Phase_1F_Research_Documentation.pdf** (14 pages) - L1 simple loss detector and L2 Sentinel-inspired reliability states
1G: **Vehicle_Tracking_Phase_1G_Research_Documentation.pdf** (12 pages) - Conditional YOLO road-vehicle candidate generation after risk/loss
1H: **Vehicle_Tracking_Phase_1H_Research_Documentation.pdf** (16 pages) - Appearance identity, M1/M2 memory, ambiguity, observation quality, selective routing
1I: **Vehicle_Tracking_Phase_1I_Research_Documentation.pdf** (14 pages) - F1-F8 failure taxonomy, manual reviews, final primary/secondary assignments

A.2 Core implementation chain

src/evaluation/event_loader.py - common frozen event contract
src/tracking/sam21_tracker.py - SAM2.1 adapter
src/tracking/dam4sam_tracker.py - DAM4SAM adapter
src/recovery/loss_detector.py - L1 loss baseline
src/recovery/l2_loss_detector.py - L2 reliability states
src/recovery/real_candidate_generator.py - real candidate plausibility filter
src/recovery/appearance_encoder.py - ResNet50 appearance encoder
src/recovery/recent_memory.py - M1 positive memory
src/recovery/distractor_memory.py - M2 distractor memory
results/phase1/phase1i_failure_taxonomy_final.csv - final 24-row taxonomy
results/phase1/phase1i_failure_summary.csv - unique-event F1-F8 summary

A.3 Minimal numeric snapshot

1F L1	TP7 / TN16 / FP0 / FN1
1F L2 risk	TP8 / TN14 / FP2 / FN0
1G retention	10/10 eligible return events
1G candidates	179 -> 146; 18.4% reduction
1H H0 / M1 / M2 Rank-1	5/10 / 5/10 / 4/10
1H exploratory R2	5 accepted, 5 correct, 0 false accepted, 50% coverage
1I F5 / F7	5 / 3 unique events

END OF PHASE 1